

- 4 The photograph shows a Baird's tapir.



Source: <https://www.biolib.cz/IMG/GAL/171566.jpg>

- (a) Baird's tapir is endemic to countries in Central America.

State what is meant by the term **endemic**.

(1)

- (b) Baird's tapir is classified as endangered.

In 2006, it was estimated that there were 5500 Baird's tapirs. This number had fallen to 3000 in 2016.

- (i) Calculate the percentage decrease in the number of Baird's tapirs from 2006 to 2016.

(2)

Answer%

- (ii) Explain how human activity, other than hunting, could have caused this decrease in the number of Baird's tapirs.

(3)

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- (c) Preservation of sperm collected from Baird's tapir may help captive breeding programmes.

Scientists investigated the effect of freezing on sperm from Baird's tapir.

The sperm were frozen and then thawed.

The results of this investigation are shown in the table.

Sperm	Percentage of sperm capable of moving (%)	Ability of sperm to swim in a straight line / a.u.	Percentage of sperm with an undamaged acrosome (%)
Freshly collected	63	3.5	80
Frozen and then thawed	38	2.5	48

(i) Describe how each of these effects of freezing could be determined.

(3)

(ii) Explain how freezing sperm could affect the success of captive breeding programmes.

(4)

(Total for Question 4 = 13 marks)